

Program overview

20-May-2025 13:14

Year 2024/2025
 Organization Mechanical Engineering
 Education Minors Mechanical Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Minors WB 2024							
WB-MI-168-24		WB-MI-168-24 Robotica					
Voor ET Bachelors							
TI3115TU	Software Engineering Methods	5					
WB3700	Robotics Minor Project	15					
WB3703	Introduction to ROS	2					
WB3705	Dynamics for Minor Robotics	4					
Voor IO Bachelors							
TI3105TU	Intermediate Python Programming	5					
WB2633	Advanced Engineering Design Project	6					
WB3700	Robotics Minor Project	15					
Voor WB Bachelors							
ET3033TU	Circuit Analysis	3					
TI3115TU	Software Engineering Methods	5					
WB3168	Robotic System Identification and Parameter Estimation	1					
WB3700	Robotics Minor Project	15					
WB3703	Introduction to ROS	2					
Voor TI Bachelors							
ET3033TU	Circuit Analysis	3					
WB3700	Robotics Minor Project	15					
WB3703	Introduction to ROS	2					
WB3704	Gazebo Assignment	2					
WB3705	Dynamics for Minor Robotics	4					
WB-MI-217-24		WB-MI-217-24 Engineering for Large-Scale Energy Conversion and Storage					
AE3516A	Fundamentals of Wind Energy I	3					
ET3034TU	Solar Energy	3					
WB3575	Energy Conversion: Devices, Systems and Efficiencies	4					
WB3580	Energy Storage	5					
WB3585	Assessment of Energy Systems	3					
WB3595	Design Project Renewables Based Energy Conversion and Storage	12					
WB-MI-238-24 Minor Authenticity and Art Crime							
MS3310UL	What is Authenticity in Art	5					
MS3320	The Biography of Art Objects: Understanding the Material Life Cycle of Artwork through Physical and	5					
MS3330EUR	Crime and Disruption in the Art Market	5					

Year	2024/2025
Organization	Mechanical Engineering
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Minors WB 2024

Year	2024/2025
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WB-MI-168-24	
Contact for students	Martin Klomp
Program Title	Robotica
In association with the Faculty of	3mE
Program Goals	The student is able to develop a functional robot in a multidisciplinary team. This comprises the following technical competences: - executing a market analysis, designing the user experience (including external appearance), producing a slick robot (IO students already possess these competences) - calculating forces, torques, stiffness, strength, acceleration/deceleration, collisions and stability of the robot, and designing the mechanical parts and the drive train (WB students already possess these competences) - calculating currents, circuit dynamics, and heat production, selecting, reading, and interpreting sensor outputs, and developing, building and testing electronic circuits for power supply, safety, and data processing (EL students already possess these competences) - designing, implementing, and testing c++/python software modules for human interfacing, interaction, safety, and autonomous behavior of the robot (TI students already possess these competences). In addition, the students will learn how to implement the knowledge from their own BSc education in a large robotics project, and they learn to negotiate and discuss interdisciplinary trade-offs.
Intended for	Students from the following five TUD bachelor educations: - Mechanical Engineering - Industrial Design - Computer Science (Technische Informatica) - Electrical Engineering - Aerospace Engineering who want to build their own robot in a multidisciplinary team, and who want to learn some of the skills of their multidisciplinary teammates.
Gives access to	This is not a schakelminor. After following this minor, you will be well-prepared for robotics-related MSc research in your own discipline. For example, after this minor, the Mechanical Engineering BSc student is recommended to take an MSc in Mechanical Engineering and work on a robotics-related thesis project.
Expected prior Knowledge	Using Linux and programming in python. This will be taught during the minor, but prior experience is useful.
Prerequisites Minor	At least 90 ECTS in one of the four BSc educations mentioned above. Additionally, strong motivation is required; admission is limited to 10 students per discipline, due to limited availability of expensive high-tech equipment and intensive coaching.
Minor Coherence / Goal	The students follow 15 ECTS of carefully selected multidisciplinary courses (no courses in your own discipline), to get acquainted with the specialization of your team members.
Minor Content 1	Each participant follows 15 ECTS of courses, and completes 15 ECTS of teamwork projects. The course list depends on your BSc background so you will not follow everything. WB3701: Design in Robotics (IDE) WB3702: Design in Robotics (ME/AE/CSE/EE) ET3033TU: Circuit Analysis (ME/AE/CSE/IDE) TI3115TU: Software Engineering Methods (ME/AE/EE) TI3105TU: Introduction to Python (IDE) WB3703: Introduction to ROS (all) WB3704: Gazebo Assignment (CSE) WB3700: Robotics Minor Project (all) WB3705: Dynamics for Minor Robotics (CSE/EE) WB2632: Project Mechanica (IDE) WB3168: Robot Mechanics (ME/AE) In addition, there are various tutorials on design tools, software packages, and production methods, all aligned with the main projects listed below (teamwork projects for all students): Robotics Minor Project During the projects you will also actively participate other activities: - (Online) meetup with coaches - (Online) meetup with customers - Participate (online) in the final demo day
Education Methods	Lectures, tutorials and projects. A team assignment is a huge part of this minor. So working in teams and flexibility is needed.
Minor Assessment	Each part is assessed separately. The minor is successfully completed if the requirements of the bachelor's programme of the student are met.
Maximum number of participants	50
Minimum number of participants	20
Minor Remarks / Schedule	NOTE: Only 50 places are available. To be able to select students in order to create balanced teams, we have temporarily put the capacity at 70 students. After you enroll, we will ask you for a motivation letter by email. Thus, successful enrollment does not guarantee admission for this minor. Descriptions of the modules and the project can be found in the digital study guide.

Year	2024/2025
Organization	Mechanical Engineering
Education	Minors Mechanical Engineering

Voor ET Bachelors

TI3115TU	Software Engineering Methods	5
Responsible Instructor	Dr.ir. M.J.G. Olsthoorn	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This course assumes you are following TI3105TU: Intermediate Python Programming at the same time as this course and will at times rely on material taught in that course. IPP asks that you complete some material before the start of their course, this course will assume you have completed this material before the start of the course as well, so make sure to go to IPP's study guide entry and complete the listed material beforehand.	
Course Contents	This course provides a comprehensive introduction to the principles and practices of software engineering. It covers the entire software development life cycle (SDLC), from requirements gathering to deployment and maintenance. Students will learn about various software development methodologies, including Agile and DevOps, and explore key concepts such as software design, testing, project management, and quality assurance. The course also addresses emerging trends in the field. Topics: 1. Introduction to Software Engineering - Definition and importance of software engineering - Overview of the software development life cycle (SDLC) - Software engineering methodologies (Waterfall, Agile, DevOps) 2. Requirements Engineering - Requirements elicitation techniques - Requirements specification and documentation - Requirements validation and management 3. Software Design - Principles of software design (modularity, abstraction, encapsulation) - Architectural design patterns - Detailed design and UML (Unified Modeling Language) 4. Software Construction - Coding standards and guidelines - Code reuse and component-based development - Code review processes and techniques 5. Software Testing - Types of testing: unit, integration, system, and acceptance testing - Test planning and design - Automated testing tools and techniques 6. Software Maintenance - Types of maintenance: corrective, adaptive, perfective, and preventive - Techniques for maintaining and evolving software - Impact of maintenance on software quality 7. Software Quality Assurance (SQA) - Quality control processes - Metrics and measurement for software quality 8. Software Configuration Management (SCM) - Version control systems (e.g., Git) - Build and release management - Change control processes 9. Emerging Trends and Technologies - DevOps methodology and CI/CD	
Study Goals	By the end of this course, students will be able to: 1. Understand the software development life cycle (SDLC). 2. Gather and analyze software requirements. 3. Design software systems using architectural and design principles. 4. Explain coding standards and best practices for software solutions. 5. Conduct various types of software testing to ensure quality and reliability. 6. Apply version control to collaborate (on source code) together.	
Education Method	Lectures + readings + assignments.	
Literature and Study Materials	Slides and material on BrightSpace.	
Assessment	The final grade of this course is calculated using the following weights: - Written exam: 100% of your final score - Assignments: Pass/Fail (you should have a pass on all assignments before you get your score) There is a single resit opportunity of the exam per academic year. The exams consist of multiple choice questions. There is a repair opportunity in case you failed one of the lab assignments. You cannot repair more than 1 lab assignment. You need to pass all lab assignments to pass the course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	This course is only accessible to students who have registered for the Computer Science minor programme (via the regular registration procedure).	
Co-Instructor	Prof.dr. A.E. Zaidman	
Co-Instructor	M. Swillus	
Co-Instructor	Prof.dr. A. van Deursen	
Co-Instructor	M. Izadi	
Co-Instructor	Dr. C.E. Brandt	
Co-Instructor	Dr. A. Panichella	

WB3700	Robotics Minor Project	15
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned.	
Study Goals	After completing this course students should be able to: - work in a multidisciplinary team - contribute, from their own discipline, to a working robotic prototype	
Education Method	The students will be coached in working together in a multidisciplinary team, contact with customers, technological problems.	
Assessment	In this course you will have multiple deliverables: a design-report (30%), the robotic prototype (40%), a handover-report (20%) and a peer review of group members (10%).	
Department	ME Department Cognitive Robotics	

WB3703	Introduction to ROS	2
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	An introduction to the Robot Operating System (ROS). This course consists of a couple of lectures followed by an assignment on a small mobile robot. In this course we will use ROS2 and the MIRTE robot. > How to use ROS communication tools (topics, services, actions) to exchange information between functional modules > Visualization and creation of a custom environment with a robot > Mapping of the robot environment and navigation with a mobile robot > How to implement a pick-and-place function with industrial robot arms > Design of a complete robotic application with state machines	
Study Goals	After completing this course you will be able to: - explain the differences of the 3 ways ROS nodes can communicate - design a full ROS2 system setup using existing relevant ROS2 packages, message types, using multiple nodes - implement a custom ROS2 node in Python3 in a custom ROS2 package	
Education Method	Lectures, followed by team assignment.	
Assessment	A final assignment will have to be handed in. This is a demo on a physical robot. The assignment will be done in pairs.	
Department	ME Department Cognitive Robotics	

WB3705	Dynamics for Minor Robotics	4
Responsible Instructor	G. Papaioannou	
Instructor	Ir. A.M. van der Niet	
Contact Hours / Week x/x/x/x	0.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Hibbeler Dynamics chapters; 12: kinematics of a particle 13: Force and acceleration (particle) 14: work and energy (particle) 15: impulse and momentum (particle) 16: planar kinematics (rigid body) 17: force and acceleration (rigid body, planar) 18: work and energy (rigid body, planar) 19: impulse and momentum (rigid body, planar) 22: vibrations	
Study Goals	LO1. Apply kinematics equation and mathematical formulations (position, velocity, acceleration notation) for the motion of point mass and of a rigid body LO2. Apply energy principles for the motion of point masses and rigid bodies. LO3. Formulate and solve the equations of motion for point masses and rigid bodies in different reference frames using vector notation. LO4. Apply the concepts of linear and angular momentum for the motion of rigid bodies in situations where impulse and collisions play a role. LO5. Analyse the motion of rigid bodies and the energy principles from data provided by lab exercises.	
Education Method	flipped classroom; watch lecture videos at home, instructional lectures at campus (not mandatory).	
Literature and Study Materials	Mandatory: 4 practice assignments, two lab reports Hibbeler Dynamics 14th edition SI units	
Assessment	No written exam; Grading based on practice assignments and lab reports.	
Department	ME Department Cognitive Robotics	

Year	2024/2025
Organization	Mechanical Engineering
Education	Minors Mechanical Engineering

Voor IO Bachelors

TI3105TU	Intermediate Python Programming	5
Responsible Instructor	F. Mulder	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	You are expected to have finished our Python Prerequisites mini-course before this course starts. Find it at https://ipp.pages.eui.tudelft.nl/python-prerequisites/	
Course Contents	This course teaches programming in Python on an intermediate level.	
Study Goals	After completion of this course, the student is able to: 1. Implement Python programs based on a given natural-language description. 2. Write code that follows Python's code style guidelines. 3. Make use of basic data structures such as strings, lists, dictionaries, tuples, and sets. 4. Implement reading and writing data from/to different file formats. 5. Apply object-oriented programming concepts such as inheritance, operator overloading, and properties. 6. Apply functional programming concepts such as list comprehensions, structural pattern matching and higher-order functions, as well as iterators and generators. 7. Understand and write type hints.	
Education Method	Seminars and lab sessions.	
Computer Use	Students do all programming assignments on their own computers (except for the exams).	
Books	Think Python 2nd Edition by Allen B. Downey (freely available from https://greenteapress.com/wp/think-python-2e/). We also provide other reading materials for topics that are not covered in this book.	
Assessment	Midterm exam (2 hours) in week 1.5, final exam (3 hours) in week 1.10. Both exams are computer exams. The midterm exam covers the course materials discussed until week 1.5, and the final exam covers all course materials. There is also a resit opportunity. The resit covers all course materials (there is no separate resit for the midterm exam). The final grade of the course is calculated as follows: $\text{course grade} = 0.2 * \max(\text{midterm}, \text{final}) + 0.8 * \text{final}$ So the midterm grade only counts if it is higher than the grade for the final exam. The final exam grade has to be at least 5.0. (And as usual, the course grade has to be at least 5.8 in order to pass the course.) The resit grade replaces the original grade only if the resit grade is greater than the original grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You may use the Python documentation during the exam.	
Enrolment / Application	This course is only accessible to students who have registered for one of these minor programmes (via the regular registration procedure): Computer Science or Robotics.	
Co-Instructor	I.D. van Kreveld	

WB2633	Advanced Engineering Design Project	6
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Course Language	Dutch	
Course Contents	wb2633	
Study Goals	see section descriptions	
Education Method	see section descriptions	
Assessment	see section descriptions	
Department	ME Department Precision & Microsystems Engineering	

WB3700	Robotics Minor Project	15
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned.	
Study Goals	After completing this course students should be able to: - work in a multidisciplinary team - contribute, from their own discipline, to a working robotic prototype	
Education Method	The students will be coached in working together in a multidisciplinary team, contact with customers, technological problems.	
Assessment	In this course you will have multiple deliverables: a design-report (30%), the robotic prototype (40%), a handover-report (20%) and a peer review of group members (10%).	
Department	ME Department Cognitive Robotics	

Year	2024/2025
Organization	Mechanical Engineering
Education	Minors Mechanical Engineering

Voor WB Bachelors

ET3033TU	Circuit Analysis	3
Responsible Instructor	Prof.dr.ir. L. Abelmann	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	The course requires familiarity with handling the complex numbers algebra.	
Course Contents	Electrical circuits fundamentals / basic concepts and components: current and charge, voltage and current sources, electrical circuits, circuit symbols, direct current, alternating current, resistors capacitors and inductors, Ohm's law, Kirchhoff's laws, power dissipation in resistors, series and parallel connections. Measurement of voltages and currents: common wave forms, measuring techniques and equipment. Resistance and DC Circuits: Thévenin and Norton theorems, superposition, nodal analysis, mesh analysis, (solving) circuit equations. Capacitance and electric fields: capacitors and capacitance, electric field strength and flux density, series and parallel connected capacitors, capacitor I/V relationship, sinusoidal voltage and current, energy stored in a capacitor. Inductance and Magnetic Fields: electromagnetism, reluctance, inductance, self-inductance, series and parallel connections, inductor I/V relationship, sinusoidal voltage and current, energy stored in an inductor, mutual inductance, transformers. Alternating voltages and currents: resistance vs. reactance, impedance, phasor diagrams, complex notation. Power in AC circuits: power dissipation in resistive circuits, power in capacitors and inductors, power in circuits with resistance and reactance, active and reactive power, power factor correction, three phase systems, power measurement. Frequency characteristics of AC circuits: two-port networks, the decibel, frequency response, filter networks, Bode diagrams, RLC circuits and resonance, stray capacitance and inductance. Transient behaviour: Charging of capacitors and energising of inductors, discharging and de-energising, first order systems, generalised response, second order systems, higher order systems.	
Study Goals	The course aims at building up insight and analysis instruments in the area of linear electrical circuits and components. It will also establish a solid foundation for understanding, analysing and synthesising electronic circuits using non-linear active components in both the analogue and digital domains.	
Education Method	Interactive lecturing as a preparation for more detailed self-study and engagement with the content. Lectures are not necessarily completely covering all topics.	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276. Material in the Brightspace learning environment.	
Assessment	Homeworks + written exam In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Permitted Materials during Tests	The exam is a closed book exam. However, a single-sided self-handwritten A4 as a cheat sheet is permitted. Only a non-programmable calculator without capabilities for storage of text or formulas and without capabilities for graphing may be used. Suitable types are the TI-30 and the Casio FX-82, and the various variants of these machines.	
Judgement	The final grade depends for 10% on the homework and for 90% on the written exam.	
maximum aantal deelnemers	100	

TI3115TU	Software Engineering Methods	5
Responsible Instructor	Dr.ir. M.J.G. Olsthoorn	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This course assumes you are following TI3105TU: Intermediate Python Programming at the same time as this course and will at times rely on material taught in that course. IPP asks that you complete some material before the start of their course, this course will assume you have completed this material before the start of the course as well, so make sure to go to IPP's study guide entry and complete the listed material beforehand.	
Course Contents	This course provides a comprehensive introduction to the principles and practices of software engineering. It covers the entire software development life cycle (SDLC), from requirements gathering to deployment and maintenance. Students will learn about various software development methodologies, including Agile and DevOps, and explore key concepts such as software design, testing, project management, and quality assurance. The course also addresses emerging trends in the field. Topics: 1. Introduction to Software Engineering - Definition and importance of software engineering - Overview of the software development life cycle (SDLC) - Software engineering methodologies (Waterfall, Agile, DevOps) 2. Requirements Engineering - Requirements elicitation techniques - Requirements specification and documentation - Requirements validation and management 3. Software Design - Principles of software design (modularity, abstraction, encapsulation) - Architectural design patterns - Detailed design and UML (Unified Modeling Language) 4. Software Construction - Coding standards and guidelines - Code reuse and component-based development - Code review processes and techniques 5. Software Testing - Types of testing: unit, integration, system, and acceptance testing - Test planning and design - Automated testing tools and techniques 6. Software Maintenance - Types of maintenance: corrective, adaptive, perfective, and preventive - Techniques for maintaining and evolving software - Impact of maintenance on software quality 7. Software Quality Assurance (SQA) - Quality control processes - Metrics and measurement for software quality 8. Software Configuration Management (SCM) - Version control systems (e.g., Git) - Build and release management - Change control processes 9. Emerging Trends and Technologies - DevOps methodology and CI/CD	
Study Goals	By the end of this course, students will be able to: 1. Understand the software development life cycle (SDLC). 2. Gather and analyze software requirements. 3. Design software systems using architectural and design principles. 4. Explain coding standards and best practices for software solutions. 5. Conduct various types of software testing to ensure quality and reliability. 6. Apply version control to collaborate (on source code) together.	
Education Method	Lectures + readings + assignments.	
Literature and Study Materials	Slides and material on BrightSpace.	
Assessment	The final grade of this course is calculated using the following weights: - Written exam: 100% of your final score - Assignments: Pass/Fail (you should have a pass on all assignments before you get your score) There is a single resit opportunity of the exam per academic year. The exams consist of multiple choice questions. There is a repair opportunity in case you failed one of the lab assignments. You cannot repair more than 1 lab assignment. You need to pass all lab assignments to pass the course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	This course is only accessible to students who have registered for the Computer Science minor programme (via the regular registration procedure).	
Co-Instructor	Prof.dr. A.E. Zaidman	
Co-Instructor	M. Swillus	
Co-Instructor	Prof.dr. A. van Deursen	
Co-Instructor	M. Izadi	
Co-Instructor	Dr. C.E. Brandt	
Co-Instructor	Dr. A. Panichella	

WB3168	Robotic System Identification and Parameter Estimation	1
Responsible Instructor	Ir. C.C. Claij	
Contact Hours / Week x/x/x/x	0/1/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Selected topics on sensors for robotics The course consists of 1-2 theoretical and 1-2 practical lectures. The theoretical lecture(s) focus on gaining knowledge on different topics and sensors: - Introduction to sensors (classification and characteristics) - Visual sensors (2D Cameras) - Depth sensors (Time-of-Flight) - Proximity sensors (IR and ultrasonic) - Beacon-based sensors (RFID and GPS) - Position sensors (Encoders) - Orientation sensors (Compass and IMU) - Force/Torque sensors - Pressure sensors - Temperature sensors - Data processing - Calibration The practical lecture(s) aim on experiencing the knowledge gained during the theoretical lectures.	
Study Goals	This course aims to provide an overview of sensors and their features for robotic applications. Next to the theoretical understanding of the use of different sensors, you will experience several of their use cases and limitations.	
Assessment	Written report	
Department	ME Department Cognitive Robotics	

WB3700	Robotics Minor Project	15
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned.	
Study Goals	After completing this course students should be able to: - work in a multidisciplinary team - contribute, from their own discipline, to a working robotic prototype	
Education Method	The students will be coached in working together in a multidisciplinary team, contact with customers, technological problems.	
Assessment	In this course you will have multiple deliverables: a design-report (30%), the robotic prototype (40%), a handover-report (20%) and a peer review of group members (10%).	
Department	ME Department Cognitive Robotics	

WB3703	Introduction to ROS	2
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	An introduction to the Robot Operating System (ROS). This course consists of a couple of lectures followed by an assignment on a small mobile robot. In this course we will use ROS2 and the MIRTE robot. > How to use ROS communication tools (topics, services, actions) to exchange information between functional modules > Visualization and creation of a custom environment with a robot > Mapping of the robot environment and navigation with a mobile robot > How to implement a pick-and-place function with industrial robot arms > Design of a complete robotic application with state machines	
Study Goals	After completing this course you will be able to: - explain the differences of the 3 ways ROS nodes can communicate - design a full ROS2 system setup using existing relevant ROS2 packages, message types, using multiple nodes - implement a custom ROS2 node in Python3 in a custom ROS2 package	
Education Method	Lectures, followed by team assignment.	
Assessment	A final assignment will have to be handed in. This is a demo on a physical robot. The assignment will be done in pairs.	
Department	ME Department Cognitive Robotics	

Year	2024/2025
Organization	Mechanical Engineering
Education	Minors Mechanical Engineering

Voor TI Bachelors

ET3033TU	Circuit Analysis	3
Responsible Instructor	Prof.dr.ir. L. Abelman	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	The course requires familiarity with handling the complex numbers algebra.	
Course Contents	Electrical circuits fundamentals / basic concepts and components: current and charge, voltage and current sources, electrical circuits, circuit symbols, direct current, alternating current, resistors capacitors and inductors, Ohm's law, Kirchhoff's laws, power dissipation in resistors, series and parallel connections. Measurement of voltages and currents: common wave forms, measuring techniques and equipment. Resistance and DC Circuits: Thévenin and Norton theorems, superposition, nodal analysis, mesh analysis, (solving) circuit equations. Capacitance and electric fields: capacitors and capacitance, electric field strength and flux density, series and parallel connected capacitors, capacitor I/V relationship, sinusoidal voltage and current, energy stored in a capacitor. Inductance and Magnetic Fields: electromagnetism, reluctance, inductance, self-inductance, series and parallel connections, inductor I/V relationship, sinusoidal voltage and current, energy stored in an inductor, mutual inductance, transformers. Alternating voltages and currents: resistance vs. reactance, impedance, phasor diagrams, complex notation. Power in AC circuits: power dissipation in resistive circuits, power in capacitors and inductors, power in circuits with resistance and reactance, active and reactive power, power factor correction, three phase systems, power measurement. Frequency characteristics of AC circuits: two-port networks, the decibel, frequency response, filter networks, Bode diagrams, RLC circuits and resonance, stray capacitance and inductance. Transient behaviour: Charging of capacitors and energising of inductors, discharging and de-energising, first order systems, generalised response, second order systems, higher order systems.	
Study Goals	The course aims at building up insight and analysis instruments in the area of linear electrical circuits and components. It will also establish a solid foundation for understanding, analysing and synthesising electronic circuits using non-linear active components in both the analogue and digital domains.	
Education Method	Interactive lecturing as a preparation for more detailed self-study and engagement with the content. Lectures are not necessarily completely covering all topics.	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276. Material in the Brightspace learning environment.	
Assessment	Homeworks + written exam In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Permitted Materials during Tests	The exam is a closed book exam. However, a single-sided self-handwritten A4 as a cheat sheet is permitted. Only a non-programmable calculator without capabilities for storage of text or formulas and without capabilities for graphing may be used. Suitable types are the TI-30 and the Casio FX-82, and the various variants of these machines.	
Judgement	The final grade depends for 10% on the homework and for 90% on the written exam.	
maximum aantal deelnemers	100	

WB3700	Robotics Minor Project	15
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned.	
Study Goals	After completing this course students should be able to: - work in a multidisciplinary team - contribute, from their own discipline, to a working robotic prototype	
Education Method	The students will be coached in working together in a multidisciplinary team, contact with customers, technological problems.	
Assessment	In this course you will have multiple deliverables: a design-report (30%), the robotic prototype (40%), a handover-report (20%) and a peer review of group members (10%).	
Department	ME Department Cognitive Robotics	

WB3703	Introduction to ROS	2
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	An introduction to the Robot Operating System (ROS). This course consists of a couple of lectures followed by an assignment on a small mobile robot. In this course we will use ROS2 and the MIRTE robot. > How to use ROS communication tools (topics, services, actions) to exchange information between functional modules > Visualization and creation of a custom environment with a robot > Mapping of the robot environment and navigation with a mobile robot > How to implement a pick-and-place function with industrial robot arms > Design of a complete robotic application with state machines	
Study Goals	After completing this course you will be able to: - explain the differences of the 3 ways ROS nodes can communicate - design a full ROS2 system setup using existing relevant ROS2 packages, message types, using multiple nodes - implement a custom ROS2 node in Python3 in a custom ROS2 package	
Education Method	Lectures, followed by team assignment.	
Assessment	A final assignment will have to be handed in. This is a demo on a physical robot. The assignment will be done in pairs.	
Department	ME Department Cognitive Robotics	

WB3704	Gazebo Assignment	2
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	In this course the student will learn about the use of the Gazebo simulator in ROS. Firstly creating your own model of the robot, and then controlling the robot, reading out sensor values, and interacting with the environment.	
Study Goals	After this course the student will be able to: - Understand the importance of a simulation of the robot. - Build a working simulation of an actual robot including sensors and actuators.	
Education Method	One 45 min lecture, followed by an assignment.	
Assessment	The final assignment of this course will be a working Gazebo simulation of a to be designed robot.	
Department	ME Department Cognitive Robotics	

WB3705	Dynamics for Minor Robotics	4
Responsible Instructor	G. Papaioannou	
Instructor	Ir. A.M. van der Niet	
Contact Hours / Week x/x/x/x	0.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Hibbeler Dynamics chapters; 12: kinematics of a particle 13: Force and acceleration (particle) 14: work and energy (particle) 15: impulse and momentum (particle) 16: planar kinematics (rigid body) 17: force and acceleration (rigid body, planar) 18: work and energy (rigid body, planar) 19: impulse and momentum (rigid body, planar) 22: vibrations	
Study Goals	LO1. Apply kinematics equation and mathematical formulations (position, velocity, acceleration notation) for the motion of point mass and of a rigid body LO2. Apply energy principles for the motion of point masses and rigid bodies. LO3. Formulate and solve the equations of motion for point masses and rigid bodies in different reference frames using vector notation. LO4. Apply the concepts of linear and angular momentum for the motion of rigid bodies in situations where impulse and collisions play a role. LO5. Analyse the motion of rigid bodies and the energy principles from data provided by lab exercises.	
Education Method	flipped classroom; watch lecture videos at home, instructional lectures at campus (not mandatory).	
Literature and Study Materials	Mandatory: 4 practice assignments, two lab reports Hibbeler Dynamics 14th edition SI units	
Assessment	No written exam; Grading based on practice assignments and lab reports.	
Department	ME Department Cognitive Robotics	

Year	2024/2025
Organization	Mechanical Engineering
Education	Minors Mechanical Engineering

WB-MI-217-24

Contact for students	Prof.dr.ir. Wiebren de Jong Section Large-Scale Energy storage (LSE) Faculty ME - Department of Process & Energy Leeghwaterstraag 39, office 34 J-0-920 (lab of Process & Energy) Telephone: 015-2789476 Mobile phone: 06-51236425 Mail: wiebren.dejong@tudelft.nl
Program Title	ELECS
Director of Education	Dr. R. Delfos Faculty ME r.delfos@tudelft.nl
EC Program	30 ECTS
Program Start (Study Year)	2019
Introduction 1	Energy conversion is the engine of the economics of our society. For a long time since the industrial revolution in the 18th century fossil fuels have dominated energy supply to a multitude of industries, the transportation sector and households. Reduction of CO2 emissions is urgently needed in view of Climate Change. Therefore, alternative energy supply is coming up rapidly. More and more, electricity will form the energy carrier for processes and household heat and power supply. Solar and wind power supply are increasingly implemented and become cheaper in fast pace. However, solar and wind are intermittent in nature and their power supply characteristics do not match the demand, neither in time nor place. There are periods and places with excess electricity supply compared to the demand, but certainly also times and locations with power supply shortage. Energy storage therefore is needed. In order to seriously reduce our societies' dependency on fossil fuels a huge effort is currently made to develop, scale-up and implement energy storage systems. Such systems are required for both short term (daily fluctuations and day-night rhythm) and long term (bridging reliable energy supply over seasons). This minor 'ELECS' covers both the aspects of renewable energy supply, energy conversion and storage and system integration
Program Goals	The aim of the minor is to equip bachelor engineering students with selected knowledge and skills concerning energy conversion and storage systems and component technologies that are based on renewable resources, which are as such mostly fluctuating and not-well predictable. Electricity is not easily and cost-effectively stored, so a mismatch both in time and geographically-between demand and supply of power is a real societal challenge. Also, storage of electricity needs to be accomplished at widely varying time scales to cope with fast weather fluctuations, day-night rhythm, occurrence of larger weather systems effects and seasonal variation of supply. This minor will deal with providing the fundamentals of a the most relevant intermittent renewable energy supply systems, wind and solar energy systems. Furthermore, systems analysis performance evaluation courses (energy storage, life cycle assessment) and practical studies (system modelling of energy conversion and storage systems and component level study design/research project) will be offered to enable students to make an evaluation of a large renewables based energy conversion and storage system. The difference with other existing minors that are offered within TU Delft is that a group of renewable energy systems is considered here in the framework of the challenge of storing energy at different time scales, taking a systems evaluation and practical component study approach together, which is unique. In summary, the topic of the minor is highly relevant due to the energy transition the world faces. Because the minor provides a comprehensive introduction to renewable energies, providing fundamental and practical knowledge and skills to students in this domain, it will be attractive and challenging to our engineering students.
Administration by the Education of	Faculty ME Mechanical Engineering
Minor Title	Engineering for Large-scale energy conversion and storage ELECS
EC Minor (Volume)	30 ECTS
Intended for	Targeted students for the Minor ELECS are those with an engineering basic knowledge of thermodynamics, heat transfer and/or fluid mechanics; i.e. students of Mechanical Engineering, Aerospace Engineering, Applied Sciences, Electrical Engineering, Technology Policy and Management, and Civil Engineering (Strictly: MT, Wb, CT, EE, TA, AE, TB, LST, MST, NB, TN)
Minor Content 1	The minor ELECS will provide students with an attractive constellation of curriculum courses and a multi-disciplinary integrated project concerning the topic of energy conversion and storage engineering. Aspects of variability in power (and heat) production characteristics and demand matching issues will be dealt with, combined with the evaluation of conversion characteristics and energy system evaluation. The undergraduate student having followed this Minor will be able to: - Understand and apply knowledge of renewable power and heat supply systems - Apply process calculations for energy conversion and storage systems based on thermodynamics, mass and heat transfer and fluid mechanics - Evaluate systems based on the concepts of exergy analysis and life cycle assessment - Provide insights into current state of the art and future trends in energy engineering - Synthesize their knowledge and competences in a final design project, including aspects of technical communication (presentations, technical report writing). Targeted students for the Minor ELECS are those with an engineering basic knowledge of thermodynamics, heat transfer and/or fluid mechanics; i.e. students of Mechanical Engineering, Aerospace Engineering, Applied Sciences, Electrical Engineering, Technology Policy and Management, and Civil Engineering (Strictly: MT, Wb, CT, EE, TA, AE, TB, LST, MST, NB, TN)
Remarks Maximum number of participants	75

AE3516A	Fundamentals of Wind Energy I	3
Responsible Instructor	Prof.dr.ir. A.C. Viré	
Instructor	Dr. F. Taruffi	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course introduces the topic of offshore wind energy by going through the main steps that a wind farm developer needs to achieve in order to build an offshore wind farm. The main topics cover in the course are: - Historical developments in wind energy - Wind energy harvesting and Betz limit - Offshore wind and wave conditions - Estimation of the energy yield - Design considerations for wind turbine farms - Economic aspects - Planning issues - Emerging concepts such as floating wind turbines	
Study Goals	At the end of this course, the students will be able to: - Explain the basic principles of wind energy conversion. - Identify the steps required when developing offshore wind farms. - Differentiate between the challenges and opportunities of offshore and onshore wind energy harvesting. - Use existing tools for estimating the power produced by an offshore wind farm and the effect of wakes. - Integrate knowledge from various fields of engineering related to the development of offshore wind parks. - Reflect on the current and future trends in the field of offshore wind energy.	
Education Method	The teaching method uses the principle of flipped classroom. This means that, every week, the students have to perform online assignments (e.g. watch a lecture, read a text, etc) prior to coming to class. In class, additional lecture materials, exercises and feedback are given to the students.	
Assessment	The final assessment is a written exam. Additional points are given for performing the online activities prior to class.	

ET3034TU	Solar Energy	3
Responsible Instructor	Prof.dr.ir. A.H.M. Smets	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In the course Solar Energy, you will learn to design a complete photovoltaic (PV) system for any application and location. This course introduces the technology that converts solar energy into electricity. The role of solar energy in both the energy transition towards a sustainable future and climate change mitigation will be discussed in detail. The physical principle of the PV energy conversion together with the design rules for a solar cell are introduced. These design rules are applied on the various PV technologies from cell up to module level. This program highlights various metrics related to performance, costs, reliability and circularity that will be used to evaluate the advantages and limitation of different PV technologies and its applications.	
Study Goals	After completion of the course, the students are able to: a) Reflect on the role of solar energy in both the energy transition towards a sustainable future and climate change mitigation b) Understand the physical working principles of photovoltaic energy conversion in solar cells c) Recognize and describe the various photovoltaic technologies and evaluate their applicability using various metrics related to performance, costs, reliability and circularity of PV. d) Design a complete photovoltaic system for any particular application and location	
Education Method	The education method is the so-called flipped-class room. Lectures will be provided in small video lectures on Brightspace. The on-campus lectures are focused on problem-solving. Students will actively work on problems that cover the entire content of the course and the various engineering skills. Solutions will be discussed during the lectures. Homework for every lecture includes: watching the relevant video lectures dedicated to the upcoming course and the remaining exercises of the previous lecture.	
Literature and Study Materials	Literature and Study Materials are provided on Brightspace: Videolectures Lectureslides. Large set of problems & exercises including answers. Answers are posted on Brightspace after the problems have been discussed in the lectures. Large set of examples of exams and answers. The book 'Solar Energy, basics, technology and systems' is free to download as e-book via all online retailers. Hardcopies of the book are sold at the ETV counter (building 36).	
Assessment	The course ET3034TU Solar Energy Energy will be assessed in Q2 and in Q3 (resit) of the academic year 2024/2025 as follows: The exam is a written exam. The exam takes 3 hours. The exam consist of open questions and multiple choice questions. A formula sheet will be provided during the exam (example is posted on Brightspace). You can use only pen, paper and a calculator. The exam will be assessed with a mark with a resolution of 0.5. To pass the exam a score equal to or higher than 58.0 out of 100.0 is needed. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Remarks	Coordinator: Arno Smets	

WB3575	Energy Conversion: Devices, Systems and Efficiencies	4
Responsible Instructor	Dr.ir. T.M.J. Nijssen	
Responsible Instructor	M. Ramdin	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic thermodynamics (1st year BSc level)	
Course Contents	This course will introduce students to the thermodynamics of energy conversion processes such as steam power plants, gas/wind turbines, electrochemical/fuel cells, and heat pumps. Thermodynamics will be applied to evaluate the efficiency and performance of energy conversion processes. In addition, devices and components used for energy conversion and storage will be discussed. The course includes the following lecturing elements: A review of basic thermodynamics, including 1st and 2nd law, energy, entropy, and exergy balances, and properties of mixtures Entropy loss of real processes Exergy loss of real processes Energy conversion CO2 Capture and conversion Solar Energy, Hydrogen and Fuel Cells Economic evaluation of energy conversion processes	
Study Goals	By the end of this course, you will be able to: - LO1: explain the working principles of energy conversion processes, systems and devices. - LO2: explain the mechanisms through which energy is lost in conversion processes. - LO3: calculate the efficiency of energy conversion processes through entropy and exergy analysis. - LO4: calculate the operational and investment costs of energy conversion processes. - LO5: assess the feasibility of an energy conversion process based on its efficiency and costs.	
Education Method	Lectures, tutorials, ungraded homework exercises and graded group assignments	
Books	Moran & Shapiro: Principles of Engineering Thermodynamics, 9th Edition	
Assessment	The final grade will be based on the group exercises (40%) and written exam (60%). The minimum grade for the written exam should be at least 5.5. Participation in the group exercises is mandatory to pass the course.	
Tags	Energy Energy & Industry Industry Process Sustainability Technology Thermodynamics exergy	
Department	ME Department Process & Energy	

WB3580	Energy Storage	5
Responsible Instructor	Prof.dr.ir. W. de Jong	
Instructor	Prof.dr. P.J. Vardon	
Instructor	Dr.ir. J.M. Bloemendal	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Energy storage becomes increasingly important in society due to the introduction of increased shares of fluctuating renewable energy sources, like wind and solar energy. Moreover, a decreased dependency on fossil fuel resources is aimed at.	
	This course will introduce the students to the different strategies and systems of energy storage (mechanical, thermal, chemical), with an emphasis on large-scale storage systems for electricity.	
	The course includes the following lecturing elements:	
	Introduction: context of renewable power supply (current and future) and scope. Mechanical storage systems (pumped hydro-electricity, compressed air energy storage, flywheels). Electrical storage systems (supercapacitors, superconducting magnetic Energy Storage). Electrochemical storage systems (batteries, flow batteries) Chemical storage systems (hydrogen production and storage options, power-to-gas, power-to-liquids, carbon sources). Carbon capture and storage/utilization. Thermal storage systems (sensible heat, phase change materials, thermo-chemical), including geothermal storage systems.	
Study Goals	The student can understand, select, and conceptually design and optimize processes for energy storage (both electricity and heat storage systems).	
	More specifically, the student must be able to: 1. understand power supply and demand characteristics in relation to intermittent renewable energy sources and the role of energy storage herein; 2. understand and select appropriate systems for energy storage in terms of capacity and power and start-up and shut-down aspects; 3. solve problems related to mechanical, thermal, thermo-mechanical, electrical and chemical energy storage components of energy storage systems based on in depth knowledge of thermodynamics, mass and heat transfer, and (fluid) mechanics; 4. design basic process features for the abovementioned classes of energy storage systems.	
Education Method	Plenary lectures including cases and solving problems (4 hours per week) and separate working classes (2 hours per week).	
Computer Use	the students have to use standard computer tools to calculate thermodynamic properties and performance of simple reaction and separation system	
Literature and Study Materials	Optional books: O.S. Burheim (2017) Engineering Energy Storage, Academic Press, 1st ed. G.Genta (1985), Kinetic Energy Storage, Butterworth R.A. Huggins (2010), Energy Storage, Springer Verlag A.F. Mills, (Basic) Heat (and Mass) Transfer, Pearson (all editions) or alike M.J. Moran, H.N. Shapiro, D.D. Boettner, and M.B. Bailey Principles of engineering thermodynamics (all editions) P.K. Nag (2014), Power Plant Engineering, McGraw-Hill, 4th ed. A.G. Ter-Gazarian (2010), Energy Storage for Power Systems, 2nd. ed. Towler and Sinnott, Chemical Engineering Design Principles, Practice and Economics, second edition. A. Vieira da Rosa (2013), Fundamentals of renewable energy processes, 3rd ed. J.W. Haverkort (2024), Electrolysers, Fuel Cells and Batteries: Analytical Modelling, DUP.	
	Articles: Ausfelder et al. (2015) Energiespeicherung als Element einer sicheren Energieversorgung, Chemie Ingenieur Technik, 87(1-2), pp. 17-89. Lopes Ferreira et al. (2013) Characterisation of electrical energy storage technologies, Energy, 53, 288-298.	
Assessment	Written exam (on-line; individual), 75% of the final grade	
	Assignment (group) on thermal energy storage systems, 25% final grade	
Permitted Materials during Tests	A standard (non programmable) calculator	
Percentage of Design	35%	
Department	ME Department Process & Energy	

WB3585	Assessment of Energy Systems	3
Responsible Instructor	Dr.ir. L. Stougie	
Instructor	Dr. L.L. Cutz IJchajchal	
Instructor	Dr.ir. G. Korevaar	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course introduces systematic ways to evaluate energy conversion and storage systems; life cycle assessment (LCA) and exergy analysis are applied to cases.	
Study Goals	1) use thermodynamic knowledge, and especially the concept of exergy, in the field of large scale energy conversion and storage systems 2) perform a system assessment with the help of energy, exergy and LCA concepts and tools 3) evaluate large scale energy conversion and storage systems and make recommendations to improve these	
Education Method	Lectures and case studies	
Literature and Study Materials	Reader	
Assessment	The exam is on campus in the form of a closed book written exam. In case on campus education and examination is not possible, the exam will be in the form of a remote written exam with several fraud prevention measures.	
Permitted Materials during Tests	Simple, non-programmable calculator	
Tags	Energy Energy & Industry Thermodynamics exergy	

WB3595	Design Project Renewables Based Energy Conversion and Storage	12
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The finalisation of the ELECS minor consists of a project (12 ECTS per student) in which the students, in interdisciplinary group work, based on the boundary conditions of a renewable energy supply system, will conceptually develop, design and evaluate an energy conversion system with storage, or will develop and test a novel energy conversion/storage concept on a component level. This can be done via modelling and/or setting up and performing experiments. The design project will take place during the 1st and 2nd period (Q1 and Q2) of the academic year. During Q1 (3ECTS) the students will more precisely define the goal of the design, do a literature search, and do preparatory work (including ordering materials, parts and sensors in case of experimental setups). Q1 will be closed with an oral presentation of the design plan. The actual design and evaluation of the process or component will take place during Q2 (9 ECTS). Q2 will be closed with a final report, a working model or prototype, and a pitch and poster presentation at the ELECS final symposium.	
Study Goals	After completion of the course, the students are able to: - apply and extend the fundamental knowledge that plays an important role in sustainable energy conversion and storage (in particular of thermodynamics, heat transfer, reaction kinetics and fluid mechanics) in designing a process. - divide a given process in sub-processes, together forming the entire process - apply thermodynamics to derive a process design of a technical system and to optimize the thermodynamic design (not for all projects). - apply heat transfer concepts to determine the main dimensions of heat exchangers (not for all projects) - apply reaction kinetics concepts to derive the main dimensions of reactors (not for all projects) - develop a list of requirements for the process design, accounting for its sustainability and limits to ethical, societal and safety - evaluate a design according to sustainability criteria which have to be formulated - write a well-structured technical report - in case of an experimental design project: deliver a working prototype - in case of a simulation design project: deliver a working model in Matlab, Python, or a programming language like C or Java - explain and defend their design in an oral (poster) presentation	
Education Method	During 14 weeks, students work in small groups of 4 to 5 students on a design project. The projects will be supervised by permanent staff of the research groups contributing to the minor. Dedicated instructions will be given on aspects of modelling and simulation and/or laboratory practices (a.o. safety).	
Assessment	In Q1, week 4, the students need to deliver a 2-to-3-page summary of the literature search. This will be evaluated by the supervisors (weight 10%). In Q1, week 9, the students need to deliver a 8-minute design plan presentation (i.e. the motivation, precise goal, results of the literature search, and an outline of their approach), followed by 3 minutes of questions. The presentations of all student groups will be organised in a joint session with as many supervisors present as possible. The presentation will be evaluated by the supervisors (weight 15%). In Q2, week 9, the students need to deliver a final report and a working model or prototype. This will be evaluated by the supervisors (report has weight 40%, model or prototype has weight 20%). A few days later in Q2, week 9, during the ELECS final symposium, the students will give a short pitch of their design in 4 minutes, followed by presentation of more details in a poster session. The pitch and poster presentation will be evaluated by the supervisors (weight 15%).	
Tags	Design Energy Energy & Industry Fluid Mechanics Group work Integrated Modelling Process Project Small groups Sustainability Technology Thermodynamics Transport phenomena Water Engineering exergy	
Department	ME Department Process & Energy	

Year	2024/2025
Organization	Mechanical Engineering
Education	Minors Mechanical Engineering

WB-MI-238-24 Minor Authenticity and Art Crime
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MS3310UL	What is Authenticity in Art	5
Responsible Instructor	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course we will study what the term authenticity means and when an artwork is considered "authentic". Also, the broader philosophical issues around the problem of forgeries will be addressed. Subsequently, students will get an introduction into current methods and strategies to detect forgeries in the field of historical objects (specifically Old Master paintings) and will gain an understanding how these research methods have developed over time. By analyzing concrete case studies, we will gain insight into the limitations and possibilities of the different research methods and nuance our ideas about originals, copies, imitations, forgeries and the different degrees of certainty in attribution issues.	
Study Goals	After this class the student will: - Understand the nuances of the different concepts of authenticity involved with authenticity research. - Analyze the challenges presented by specific art/cultural heritage objects by interpreting different types of knowledge sources. - Define problem definitions and relevant focal points for a specific case study.	
Education Method	Lectures and on-site museum visit	
Assessment	Portfolio, on an art object in a public collection	
Department	ME Department Materials Science & Engineering	

MS3320	The Biography of Art Objects: Understanding the Material Life Cycle of Artwork through Physical and	5
Responsible Instructor	Dr. M.W.E.M. Alfeld	
Instructor	Prof.dr. J. Dik	
Instructor	Dr.ir. D.J.M. Ngan-Tillard	
Instructor	Dr. L. van Eijck	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	MS3350: Authenticity and Art Crime: A Challenging Case Study	
Expected prior knowledge	The basics of chemistry and physics will be briefly repeated in the class, so that the willingness to learn is required but no formal training in the fields of science or engineering.	
Course Contents	An artwork is often valued primarily on account of its attribution to a specific creator or location of origin, date or period, and provenance, i.e. not only on the basis of its physical appearance. While an attribution may provide an historical context to the work, establish quality and uniqueness, all these subjectively meaningful parameters are undermined should the work turn out to be a forgery. When authenticating art, experts eventually decide what is of cultural significance and what is not. Authenticity has concrete repercussions on the market, in the form of its influence on an artworks economic value, and in law, given the liability of experts and sellers for misattributions and the sale of forgeries. For the past two decades the art world has increasingly relied on forensics, i.e. the dedicated chemical and physical investigation of artwork to help answer questions of authenticity. Nowadays, the public museum field, but also the commercial markets such as auction houses and international art fairs involve forensic analysis in their vetting procedure. The expert opinion of the art expert (connoisseurship), the line of ownership of an artwork (the provenance), and materials analysis (forensics) are currently the three most important pillars to establish the authenticity of an artwork. The materials of artworks offer a unique insight into the making of art, covering the entire lifecycle of an object. Raw materials, their geographic and chronological origin, artist's working techniques, material decay, restoration treatments and re-use and recycling of art objects are all part of the artwork lifecycle. Understanding these aspects is key to understanding the artwork's biography, its provenance and, ultimately, the artwork's authenticity. The fundamentals of this life cycle will be presented in an introductory lecture and later illustrated throughout this course in case studies. The main focus of the introduction will be on paint and painted objects and the methods used for the investigation of historical paintings (UV, IRR, XRR). The tremendous methodological progress in non-destructive analysis and chemical and physical imaging diagnostics achieved in the past decades will be illustrated in a series of lectures, which take the variable entry level of the students into account. The lectures will start from the nature of matter (atoms, their cores and electron shells) and radiation, especially X-rays. This basic instruction will be consolidated by peer instruction in the project groups. The production and detection of X-rays and their interaction with matter will be presented and it will be explained how these are used to acquire computed tomography data. The value this non-invasive 3D technique has to gain insight into an objects lifecycle will be illustrated on pottery/ clay cuneiform tablets, stone (beads/ flints), glass (flasks and beads) and corroded metal (coins/ filigree sphere) objects. X-ray spectroscopy and diffraction will be discussed at a basic level and the use of these basic principles in MA-XRF and MA-XRD imaging will be illustrated in studies of pigments of 17th century paintings and archaeological objects from classical antiquity (Etruscan and Greek sites). The individual techniques will be concluded with an introduction to neutrons and their use in neutron radiography, tomography, gamma spectroscopy and neutron activation analysis with application examples. This series of lectures will be closed by two events: the students will visit the TU Delft campus and travel in groups to see instruments that are used for the techniques described above. Finally, in a formative test the student groups will be prepared for the final exam.	
Study Goals	After this course the student is capable to: Analyze the challenges presented by potential forgeries by interpreting different knowledge sources. Discuss the physical and chemical principles behind common analysis techniques and the build-up of artworks. Formulate the materials science questions for an art object. Propose suitable methods for investigating these questions. Interpret the acquired data to contribute to an answer in the discussion of a potential forgery.	
Education Method	Lectures Site and Laboratory visits	
Assessment	The grades of the students will be derived from a closed book, written exam.	
Department	ME Department Materials Science & Engineering	

MS3330EUR	Crime and Disruption in the Art Market	5
Responsible Instructor	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Crime and Disruption in the Art Market is part of the Leiden-Delft-Erasmus Joint Minor Authenticity and Art Crime. Methods, Materials, and the Market. It will provide students with an overview of how criminal actors can disrupt the art market and how the market itself can become a multi-dimensional criminogenic space. A range of criminal activities are discussed such as forgery, theft, looting, and fraud and their impact on collecting and the art market. The topic is approached from the fields of cultural economics, cultural sociology, and critical heritage studies, providing a transdisciplinary perspective.	
Study Goals	After completion of this course, students are able to: - Describe the art market, its seminal actors, structure, and operation. -Describe the role of intermediaries and their challenges related to art quality and authorship. -Analyse why authenticity has become so important in the art market and what (criminal) implication this has. -Examine how looting, theft, and forgeries impact the art market. -Examine how society perceives cultural heritage plagued by art crime.	
Education Method	Seminars	
Assessment	Group presentation (30%) Individual final paper (70%) Attendance (mandatory component)	
Department	ME Department Materials Science & Engineering	

Prof.dr.ir. L. Abelmann

Department	Bio-Electronics
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